

IN DEPTH

REPL

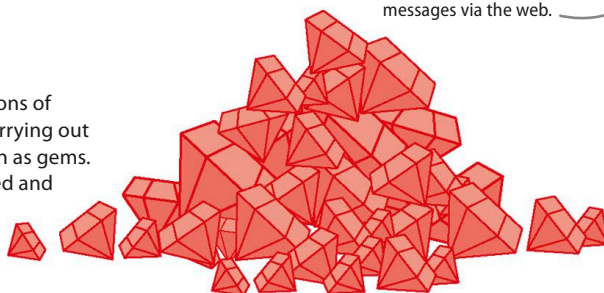
The Ruby interactive interpreter is an example of a REPL, a Read-Evaluate-Print Loop. This is a program that takes one command at a time, runs it, and prints the result. REPLs are a common feature of interpreted languages, which are often referred to as scripting languages. The quick feedback means it can be a useful tool for learning a language.

What is Ruby used for?

Ruby was made popular by “Ruby on Rails”, a framework for making websites. It allows programmers to create websites that are connected to a database. The Rails framework simplifies retrieving and displaying data from the database, and allows users to input data through the website. Rails combines Ruby with the languages of web programming: HTML, CSS, and JavaScript.

▷ **RubyGems**

Ruby’s libraries, collections of ready-made code for carrying out specific tasks, are known as gems. They can be downloaded and installed using RubyGems, Ruby’s built-in tool for gem management.



This is used for spotting and fixing spelling errors.

This sends secure messaging via the web.

```
bigdecimal (default: 1.3.0)
did_you_mean (1.1.0)
io-console (default 0.4.6)
json (default: 2.0.4)
minitest (5.10.1)
net-telnet (0.1.1)
openssl (default: 2.0.5)
power_assert (0.4.1)
psych (default: 2.2.2)
rake (12.0.0)
rdoc (default: 5.0.0)
test-unit (3.2.3)
xmlrpc (0.2.1)
```

List of local gems

Why use Ruby?

Ruby was designed to reflect how people think about problems, rather than how computers think about them. All high-level languages abstract away from the way computation is done by a computer’s hardware. Ruby is particularly focused on this approach, making it easy for users to program in a variety of styles.

REAL WORLD

Sonic Pi

A free program, Sonic Pi was built using Ruby. It turns a computer into a musical instrument, which can be played by typing code. Most of its commands are specially created to allow users to do musical tasks, but it uses many of Ruby’s basic features as well.

```
use_synth :piano
8.times do
  play :c4
  sleep 0.5
  sample :drum_cowbell
  sleep 0.5
end
```

Advantages	Disadvantages
Commands are often closer to a human language than other languages.	Used in fewer areas in the programming world than other languages.
Thanks to Ruby on Rails, Ruby is one of the fastest-growing programming languages.	Ruby code runs slower than compiled languages like Java or C.
Ruby is still being actively developed, keeping pace with new technologies.	There are fewer libraries for coding in areas other than the web.